

flea

NEWS 92 *November 2023*



<http://esanetworks.org/>

FLEA NEWS is a twice-yearly newsletter about fleas (Siphonaptera). Before including in publications any of the citations given here, please review for accuracy and correct formatting. Many of our sources are abstracting journals and current literature sources, and editors have not necessarily checked citations.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Citations for new flea species descriptions, Flea research citations from journals or conference proceedings that are not available or indexed, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

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Editorial

"Flea eggs"

Dear *Flea News* Reader,

"However, our knowledge of most of the developmental stages in the order [Siphonaptera] is so meager and incomplete as to be of little more than passing interest at this point."

R.E. Lewis (1998)

Entomologists have described the eggs of only a score of flea species from what I have gathered (Table 1). Many of these descriptions are cursory, or without scanning electron micrography.

Flea eggs have the advantage of being simple morphologically when compared with adult fleas, and when squashed onto slides they can be used to count chromosomes.

Lewis (1998) reviewed the subject of eggs and other immature flea stages briefly. Pilgrim (1993) listed Chinese literature on flea larvae, but wrote "If there is sufficient interest, a complementary list of papers referring to other aspects of developmental biology and immature stages of fleas will be offered.", but to my knowledge this list was never published.

Several authors noted various characters of eggs with possible significance to physiological ecology and phylogeny such as egg size, development time, permeability, aeropyles, chorion and extrachorion surface reticulation and hardness, and micropyle numbers.

For examples, *S. tripectinata* eggs appear hard and impermeable (Rothschild 1975), but cat flea eggs appear "delicate" and permeable (Marchiondo et al. 1999). Lateral aeropyles are likely involved with water balance and gas exchange (Marchiondo et al. 1999). They seemingly tend to show up not in evolutionarily old flea groups (*C. minerva*), but in new ones (Ceratophyllidae) (Linley et al. 1994).

Yours in fleas,
 R.L. Bossard
 Editor, *Flea News*

Table 1. Species of flea whose eggs have been examined.

	Stephanocircidae
Craneopsylla minerva	Linley et al. 1994

Sphinctopsylla ares Krasnov 2008, cit. in Ezquiaga & Lareschi 2012

Hystrihopsyllidae

Conorhinopsylla stanfordi Linley et al. 1994
Epitedia faceta Linley et al. 1994
Hystrihopsylla spp. Chen & Wang 1993, cit. in Lewis, R.E. 1998,
 Krasnov 2008, cit. in Ezquiaga & Lareschi 2012
Hystrihopsylla dippiei cit. in Kessel 1939
Stenoponia tripectinata Rothschild 1975

Malacopsyllidae

Malacopsylla grossiventris Ezquiaga & Lareschi 2012
Phthiropsylla agenoris Ezquiaga & Lareschi 2012

Pulicidae

Ctenocephalides felis Kessel 1939, Linley et al. 1994, Marchiondo et al. 1999)
Echidnophaga oshanini Vashchyonok 1966
Xenopsylla cheopis Bai 1972
Xenopsylla ostia Bai 1972

Ischnopsyllidae

Ischnopsyllus octactenus Rothschild et al. 1986
Sternopsylla distincta Linley et al. 1994

Ceratophyllidae

Aetheca wagneri Linley et al. 1994
Nosopsyllus fasciatus cit. in Kessel 1939
Orchopeas leucopus Linley et al. 1994

Literature

Bai, M. K. 1972. Studies on host-flea relationship. Part I. Histochemistry of egg yolk in *Xenopsylla cheopis* (Roths) and *Xenopsylla ostia* (Roths). Indian J. Med. Res. 60:752-56, cit. in Rothschild 1975.

Chen, J.L. and D.Q. Wang. 1993. Scanning electron microscopy of the chorion surface of three species of domestic rat fleas in south China (Siphonaptera). Endem. Dis. Bull. 8: 17-19.

Chen, J. L., and D. Q. Wang. 1993. Comparative morphology of rodent flea eggs in China. Medical and Veterinary Entomology 7: 384–386.

Ezquiaga, M.C.; Lareschi, M. Surface ultrastructure of the eggs of *Malacopsylla grossiventris* and *Phthiropsylla agenoris* (Siphonaptera: Malacopsyllidae). J. Parasitol. 2012, 98 (5), pp. 1029-1031.

Kessel, E. L. 1939. Embryology of fleas. Smithsonian Misc. Coll. 98(3): 78.

Lewis, R.E. 1998. Résumé of the Siphonaptera (Insecta) of the world. J. Med. Entomol, 35(4): 377-389.

Linley, J. R., A. H. Benton, and J. F. Day. 1994. Ultrastructure of the eggs of seven flea species

(Siphonaptera). J. Med. Entomol. 31: 813-827.

Marchiondo, A.A., S.M. Meola, K.G. Palma, J.H. Slusser & R.W. Meola. 1999. Chorion formation and ultrastructure of the egg of the cat flea (Siphonaptera: Pulicidae). Journal of Medical Entomology 36(2): 149-157.

Pilgrim, R.L.C. 1993. An annotated bibliography of the publications on flea larval morphology in Chinese periodicals, 1956-1992 (July). Flea News 46: 511-515.

Rothschild, M. Recent advances in our knowledge of the order Siphonaptera. Ann. Rev. Entomol. 1975, 20, pp. 241-259.

Rothschild, M., Y. Schlein, and S. Ito. 1986. A colour atlas of insect tissue via the flea. Wolfe, London.

Vashchyonok, V. S. 1966. Histological characteristics of oogenesis in fleas *Echidnophaga oshanini* Wagn. during feeding and reproduction. Entomol. Obozr. 45(2): 715-275, cit. in Rothschild 1975.

Announcements

Lewis World Flea Species List

Updated and available online at Catalogue of Life Checklist, GBIF (Global Biodiversity Information Facility):

Citation: Hastriter, M. W., & Bossard, R. L. (2023). Robert E. Lewis World Species Flea (Siphonaptera) List. In: O. Bánki, Y. Roskov, M. Döring, G. Ower, D. R. Hernández Robles, C. A. Plata Corredor, T. Stjernegaard Jeppesen, A. Örn, L. Vandepitte, D. Hobern, P. Schalk, R. E. DeWalt, M. Keping, J. Miller, T. Orrell, R. Aalbu, J. Abbott, R. Adlard, E. M. Adriaenssens, et al., Catalogue of Life Checklist (Version 2023-06-22). doi: <https://doi.org/10.48580/dfs-48f>

Susceptible and resistant bedbug colonies available

"The USDA, ARS, Invasive Insect Biocontrol and Behavior Lab has completed all planned research with bed bugs and we can no longer continue to rear these colonies. We have a colony of insecticide susceptible bed bugs (*Cimex lectularius*) established from bugs originally obtained from Harold Harlan (Crownsville, MD, USA) (the so-called "Harlan strain"), and a colony of pyrethroid resistant *C. lectularius* established from bugs originally obtained from the University of Massachusetts. Instead of destroying these, we would prefer to find colleagues who could use them for research purposes to whom we could transfer the strains.

If anyone is interested in receiving these colonies of susceptible and resistant bed bugs for research purposes, please get in touch and we can arrange to ship them to you.

These colonies will be destroyed if no one reaches out to request them."

Contact: Dawn Gundersen-Rindal, Research Leader Invasive Insect Biocontrol and Behavior Laboratory, USDA Agricultural Research Service Beltsville, MD 20705 ✉ dawn.gundersen-rindal@usda.gov

Featured Research

New Flea Species 2023

Eckerlin R.P., Durden L.A.

A NEW SPECIES OF FLEA (SIPHONAPTERA: CTENOPHTHALMIDAE) PARASITIZING VOLES AT HIGH ELEVATIONS IN THE GREAT SMOKY MOUNTAINS OF TENNESSEE AND NORTH CAROLINA.

(2023) The Journal of parasitology, 109 (2), pp. 107 - 112
DOI: 10.1645/22-85

Both sexes of *Catallagia appalachiensis* n. sp. are described from high elevation spruce-fir forests in Sevier County, Tennessee and adjoining Swain County, North Carolina in the Great Smoky Mountains National Park. The type host of the new flea is the southern red-backed vole, *Myodes gapperi* (Vigors) (25 flea specimens), although small numbers of specimens were also collected from a sympatric northern short-tailed shrew, *Blarina brevicauda* (Say) (2 fleas), a red squirrel, *Tamiasciurus hudsonicus* (Erxleben) (1 flea), and a North American deer mouse, *Peromyscus maniculatus* (Wagner) (1 flea). Infestation prevalences for these hosts are provided. The new species is compared morphologically with other known species of *Catallagia*, in particular with *Catallagia borealis*, the only other described congeneric flea in eastern North America. This is the first new species of flea to be described from the eastern United States since 1980.

[*Catallagia appalachiensis* Eckerlin & Durden 2023]

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Hastriter M.W.

Elevation of *Spilopsyllus inaequalis interrupta* (Jordan, 1925) (Insecta, Siphonaptera) to Full Specific Status Based on Morphology and Geographical Distribution.

(2023) Annals of Carnegie Museum, 88 (1), pp. 1 - 12
DOI: 10.2992/007.088.0101

Spilopsyllus inaequalis inaequalis and *Spilopsyllus inaequalis interrupta* are common ectoparasites of species of Leporidae. Kohls (1940) suggested that there are intergrades of these subspecies in sympatric areas of southeastern Washington and Oregon, south central Idaho, and Utah. Kohl's conclusions were based on traditional morphological features of the "clasper" (basimere and telomere) and chaetotaxy of the outer surface of the hind tibia. These morphological features, and that

of the aedeagus, have never been critically studied until now. The aedeagi of both taxa were examined and several consistent structures separate these closely allied species. The median dorsal lobe and apex of the sclerotized inner tube show little variation, although they are distinctly different among the sympatric populations and are consistent in peripheral populations. These characters are stable with little variation and their structure appears to be sufficiently different to potentially inhibit interbreeding that typically occurs where subspecies meet. These characters are valid distinguishing characters even when males of both species occur simultaneously on the same host. These newly recognized static characters of the aedeagus are illustrated. The lateral surface of the hind tibia is the most valid criterion for separation of females of the two species, although this cannot be determined with any degree of certainty. *Spilopsyllus inaequalis interrupta* is herein elevated to full specific status (*Spilopsyllus interruptus*) (Amended Status). Table 1 provides a resource for the terminology of various lobes and processes of the telomeric appendages of males of the superfamily Pulicoidea. Host and distribution records available in the literature are summarized in Tables 2 and 3.

[*Spilopsyllus interruptus* (Jordan, 1925)]

Taxonomy

Hastriter M.W.

An Assemblage of Fleas (Siphonaptera) From Canada, México, and the United States of America, in the Hastriter Collecton (Brigham Young University).

(2023) Annals of Carnegie Museum, 88 (1), pp. 13 – 90, DOI: 10.2992/007.088.0102

An accumulation of fleas that constitute the Hastriter flea collection includes accounts/records of 167 species over 48 years. These officially belong to the Brigham Young University flea collection. They are summarized in this treatise with identifications of new state, county, and provincial records from the United States, Canada, and some species that extend south into México. Records included are represented by eight families that are presented in alphabetical order (Ceratophyllidae, Ctenopthalmidae, Hystrichopsyllidae, Ischnopsyllidae, Leptopsyllidae, Pulicidae, Rhopalopsyllidae, and Vermipsyllidae). Major synonymies are provided with each species (excluding very common and cosmopolitan species) with major political subdivisions following each reference/pagination. Two appendices are included that list important references (not included in synonymies) (Appendix I) and a listing of synonyms for the family Ceratophyllidae (Appendix II). Four tables are included that list new geographical records and one table that notes new host records. Seventy-eight flea species (some of the same species were new for different state, county and provincial records) accounted for 129 new U.S. state or Canadian provincial records. These also included 12 new state records. An additional 15 new host records for 13 different flea species were documented. To aid future investigations, significant research findings such as disease investigations, zoonotic disease control measures, seasonal bionomics, et cetera are annotated in the remarks.

Hosts and Distribution

SANCHEZ J.P., LÓPEZ BERRIZBEITIA F., EZQUIAGA M.C. 2023.

Host specificity of flea parasites of mammals from the Andean Biogeographic Region.

Medical and Veterinary Entomology 37(3): 511-522.

doi: 10.1111/mve.12649.

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Bejec G.A., Bucol L.A., Ancog A.B., Pagente A.C., Panerio J.J.M., Bejec A.L.N., Belanizo J.D., Tuastomban D.J.S., Jose R.P.

Diversity of bat ectoparasites from the caves of selected Key Biodiversity Areas in Central Visayas, Philippines.

(2023) Biodiversitas, 24 (3), pp. 1693 – 1703.

Bats harbor numerous species of ectoparasites, such as ticks and mites. Many bat ectoparasites are implicated as vectors of pathogens ranging from bacteria to protozoa and viruses. Recent studies verified bats as a natural reservoir of multiple zoonotic viruses, including Middle East Respiratory Syndrome (MERS) and Severe Acute Respiratory Syndrome (SARS) coronaviruses. This study assessed the composition, diversity, and prevalence of ectoparasites of cave-dwelling bats in the four Key Biodiversity Areas (KBAs) of Central Visayas. A total of 20 species of bats were recorded in the 30 surveyed caves. Of these, 30 species of ectoparasites belonging to eight families (dipteran flies Streblidae and Nycteribiidae, mites Spinturnicidae, Macronyssidae and Trombiculidae, ticks Argasidae and Ixodidae, and Ischnopsyllidae fleas) were collected and identified. *Leptotrombidium* sp. had the highest number of individuals (n=1,684) harbored in 12 bat host species. The highest prevalence (100%) was noted in four bat host species: Philippine pygmy fruit bat *Haplonycteris fischeri* Lawrence 1939, Philippine forest roundleaf bat *Hipposideros obscurus* Peters 1861, Philippine pygmy roundleaf bat *Hipposideros pygmaeus* Waterhouse 1843, and Common bent-winged bat *Miniopterus schreibersii* Kuhl 1817. The lowest prevalence was noted in the Round-eared tube-nosed bat *Murina cyclotis* Dobson 1872, where no ectoparasite was collected. The highest intensity (n=65) of ectoparasites was observed in the Common Asian ghost bat *Megaderma spasma* Linnaeus 1758. This study provides essential data for future reference in monitoring bat population status and conservation efforts in the region. Given the close relationship between the local human community and bats (e.g., hunting and consumption), more work is needed to address the potential pathogen risks from zoonotic transmission from bats and ectoparasites.

[Editor's comment: Two species of flea were found, *Thaumapsylla breviceps* Rothschild 1907, and an unidentified *Ischnopsyllus* species, at low densities compared to other ectoparasites there (i.e. nycteribiids, streblids, mites, and ticks). Only 1.12% of bats examined had fleas.

This rarity is consistent with previous reports on bat fleas, with tropical areas seeming to have fewer bat fleas than temperate areas, and certain bat caves seeming to lack fleas altogether, even though the environmental conditions seem amenable. It seems possible that predation or hyperparasitism is diminishing bat flea numbers.]

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Bossard, Robert L., Marcela Lareschi, Mara Urdapilleta, Cristina Cutillas, and Antonio Zurita. 2023. Flea (Insecta: Siphonaptera) family diversity. Diversity 15, no. 10: 1096.

<https://doi.org/10.3390/d15101096>

This overview of extant Siphonaptera lists 19 families with major hosts and their general distribution, estimated numbers of genera, species, and subspecies, with a brief taxonomic and phylogenetic review. With around 10 new species described annually, extant flea fauna comprises an estimated 249 genera, 2215 species, and 714 subspecies globally, mostly mammal parasites, but 5% of species are on birds. Host specificity varies from euryxenous (i.e., infesting two or more host orders) (e.g., cat fleas, *Ctenocephalides felis felis*) to monoxenous (e.g., rabbit fleas, *Spilopsyllus cuniculi*). The largest family is the paraphyletic Hystrichopsyllidae, making up a third of all flea species. The largest monophyletic family, Ceratophyllidae (rodent and bird fleas), comprises another 20% and has dispersed to every continent, including Antarctica. Fleas descend from scorpionflies (Mecoptera), possibly snow scorpionflies (Boreidae) or Nannochoristidae, and even giant fossils found from the Mesozoic could be Siphonaptera. The diversification of fleas shows evidence of taxon cycles. “Relict” families, such as helmet fleas (Stephanocircidae), have a disjunct distribution reflecting the breakup of Gondwanaland 70 million years ago. “Niche specialists” include nest fleas (*Anomiopsyllus*) [sic, *Anomiopsyllus*], bat fleas (Ichnopsyllidae), and burrowing fleas, such as chigoes (Tungidae). By contrast, Ceratophyllidae fleas could be considered “great speciators”. Cat fleas and several other synanthropic flea species are invasive “supertramps”. Although those species are intensively studied, many flea species and their hosts require urgent surveys and conservation.

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Karbowiak, G.; Stanko, M.; Smahol, K.; Werszko, J.; Rychlik, L.

Parasitic Arthropods of Soricinae Shrews in North-Eastern Poland.

Animals 2023, 13, 2960. <https://doi.org/10.3390/ani13182960>

Simple Summary: Studies on the ectoparasites of insectivores are quite rare compared with other groups of mammals. The study of the ectoparasite fauna of the insectivores: *Sorex araneus*, *Sorex minutus*, *Neomys fodiens*, and *Neomys anomalus*—was carried out in three locations in north-eastern Poland: Białowież'a National Park, Kosewo Górne in the Masurian Lake District, and one location in vicinity of Warsaw. Three species of ixodid ticks, eleven species of fleas and four species of mites from the order Mesostigmata were noted, in comparison to the total number of ectoparasite species on Soricinae, recorded in Central Europe, amounts to eighty-one species: six tick species, twenty-three gamasid mites, nine trombiculid mites, nine Myobiidae and Sarcoptiformes, twenty-nine flea species, and five sucking lice. The most numerous ectoparasites recorded during the study are ticks *Ixodes ricinus* (larvae), *Dermacentor reticulatus* (nymphs), fleas *Palaeopsylla soricis*, *Megabothris walkeri*, and *Hystrichopsylla orientalis*. These species show the highest prevalence and show the highest dominance index. The parasitofauna of *S. araneus* is much richer in species than other insectivorous species. The structure and dominance of parasite assemblages differ between locations.

[The authors found "11 species of fleas: *Palaeopsylla soricis* (Dale, 1878), *Palaeopsylla similis* Dampf, 1910, *Ctenophthalmus agyrtes* (Heller, 1896), *Ctenophthalmus assimilis* (Taschenberg, 1880), *Ctenophthalmus bisoctodentatus* Kolenati, 1863, *Ctenophthalmus uncinatus* (Wagner, 1898), *Peromyscopsylla bidentata* (Kolenati, 1860), *Megabothris turbidus* (Rothschild, 1909), *Megabothris walkeri* (Rothschild, 1902), *Hystrichopsylla orientalis* Smit, 1956, *Doratopsylla dasyncnema* (Rothschild, 1897)".

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Li H., Liang Y., Dong L., Li C., Zhang L., Wang B., Ma D., Mu Q., Wang J., Hou H., Liu Q.

Predicting global potential distribution of *Peromyscopsylla hesperomys* and *Orchopeas sexdentatus* and risk assessment for invading China under climate change.

(2023) *Frontiers in Public Health*, 10, art. no. 1018327,

DOI: 10.3389/fpubh.2022.1018327

Background: *Peromyscopsylla hesperomys* and *Orchopeas sexdentatus* are regarded to be representative plague vectors in the United States. The incidence of plague is rising globally, possibly due to climate change and environmental damage. Environmental factors such as temperature and precipitation have a significant impact on the temporal and spatial distribution of plague vectors.

Methods: Maximum entropy models (MaxEnt) were utilized to predict the distributions of these two fleas and their trends into the future. The main environmental factors influencing the distribution of these two fleas were analyzed. A risk assessment system was constructed to calculate the invasion risk values of the species.

Results: Temperature has a significant effect on the distribution of the potentially suitable areas for *P. hesperomys* and *O. sexdentatus*. They have the potential to survive in suitable areas of China in the future. The risk assessment system indicated that the risk level for the invasion of these two species into China was moderate.

Conclusion: In order to achieve early detection, early interception, and early management, China should perfect its monitoring infrastructure and develop scientific prevention and control strategies to prevent the invasion of foreign flea vectors.

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Mays Maestas S.E., Campbell L.P., Wisely S.M., Dingman P.A., Reeves L.E., Kaufman P.E.

Comparison of ectoparasite communities of sylvatic and urban wild mesomammals and unowned community cats in north-central Florida.

(2023) *Journal of Medical Entomology*, 60 (3), pp. 460 - 469,

DOI: 10.1093/jme/tjad026

The adaptation of wildlife species to urban environments can drive changes in the ecology of ectoparasites and vector-borne disease. To better understand ectoparasite dynamics in an urban environment, we investigated the ectoparasite communities of 183 sylvatic and urban opossums and raccoons captured across four seasons at a rural research station and within the city of Gainesville, FL, and of 115 community cats from the Gainesville, FL area. *Amblyomma americanum* (L.) (Acari: Ixodidae), *Dermacentor variabilis* (Say), and *Ixodes texanus* Banks were collected from raccoons, *A. americanum*, *D. variabilis*, and *Ixodes scapularis* Say from opossums, and *A. americanum* from cats. Few ticks were collected from urban animals, although species richness of ectoparasites was similar between urban and sylvatic habitats. *Ctenocephalides felis* (Bouché) (Siphonaptera: Pulicidae) was collected from all sampled host species, but was particularly abundant on opossums. Additionally, *Orchopeas howardi* (Baker) (Siphonaptera: Ceratophyllidae) was collected from raccoons, and *O. howardi* and *Xenopsylla cheopis* (Rothschild) (Siphonaptera: Pulicidae) from opossums. Only raccoons were infested with raccoon lice, and only cats were infested with cat lice. Primarily opossums were infested with mites. Ectoparasite community composition varied by habitat, host species, and season; seasonal variation in ectoparasite communities differed between the sylvatic and urban habitats. While

urban mesomammals did not appear to play an important role in supporting tick populations in an urban habitat, urban opossums appear to serve as an alternate host for large numbers of cat fleas, which may be an important consideration for treatment and control efforts against ectoparasites of companion animals.

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Smith A.T., Krasnov B.R., Horak I.G., Ueckermann E.A., Matthee S.

Ectoparasites associated with the Bushveld gerbil (*Gerbilliscus leucogaster*) and the role of the host and habitat in shaping ectoparasite diversity and infestations

(2023) Parasitology, 150 (9), pp. 792 - 804,

DOI: 10.1017/S0031182023000562

Rodents are known hosts for various ectoparasite taxa such as fleas, lice, ticks and mites. South Africa is recognized for its animal diversity, yet little is published about the parasite diversity associated with wild rodent species. By focusing on a wildlife-human/domestic animal interface, the study aims to record ectoparasite diversity and levels of infestations of the Bushveld gerbil, *Gerbilliscus leucogaster*, and to establish the relationship between ectoparasite infestation parameters and host- and habitat factors. Rodents (n = 127) were trapped in 2 habitat types (natural and agricultural) during 2014-2020. More than 6500 individuals of 32 epifaunistic species represented by 21 genera and belonging to 5 taxonomic groups (fleas, sucking lice, ticks, mesostigmatan mites and trombiculid mites) were collected. Mesostigmatan mites and lice were the most abundant and fleas and mesostigmatan mites the most prevalent groups. Flea and mesostigmatan mite numbers and mesostigmatan mite species richness was significantly higher on reproductively active male than female rodents. Only ticks were significantly associated with habitat type, with significantly higher tick numbers and more tick species on rodents in the natural compared to the agricultural habitat. We conclude that the level of infestation by ectoparasites closely associated with the host (fleas and mites) was affected by host-associated factors, while infestation by ectoparasite that spend most of their life in the external environment (ticks) was affected by habitat type.

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Urdapilleta M., Lamattina D., Burgos E.F., Salomón O.D., Lareschi M.

Specificity of fleas associated with opossums in a landscape gradient in the Paranaense Rainforest Ecoregion.

(2023) Zootaxa, 5264 (4), pp. 579 - 586,

DOI: 10.11646/zootaxa.5264.4.7

The degree of host specificity of fleas varies from highly specific (monoxenous) to opportunistic (polyxenous). Specific parasite-host associations can be observed among some flea families and tribes and mammalian orders, such as fleas of the tribe Tritopsyllini (Ctenophthalmidae, Doratopsyllinae) and opossums (Didelphimorphia). Specimens of the family Didelphidae are common hosts of fleas of the genus *Adoratopsylla* (Ewing, 1925). In northern Argentina, *Adoratopsylla* (*Adoratopsylla*) *antiquorum antiquorum* (Rothschild, 1904) and *Adoratopsylla* (*Tritopsylla*) *intermedia intermedia* (Wagner, 1901) were recorded preferentially parasitizing opossums. In order to study parasite/host relationships, fleas were collected from opossums captured in different environments in

the Paranaense Rainforest ecoregion, northern Misiones province between 2016 and 2018. A total of 287 fleas were collected from 110 opossums. The fleas were identified as Pulicidae: *Ctenocephalides felis felis* (Bouché, 1835); Rhopalopsyllidae: *Polygenis (Polygenis) rimatus* (Jordan, 1932), *Polygenis (Polygenis) roberti roberti* (Rothschild, 1905); Ctenophthalmidae: *A. (T.) i. intermedia*, *A. (A.) a. antiquorum*, *Adoratopsylla (Adoratopsylla) antiquorum ronnai* (Guimarães, 1954). We report for the first time in Argentina a male specimen of *A. (A.) a. ronnai* collected on *Didelphis albiventris* (Lund, 1840) (Didelphidae), and male and female specimens of *P. (P.) r. roberti* collected on *Didelphis aurita* (Wied-Neuwied, 1826), and we describe the expansion of the geographic and host distribution of fleas to the Paranaense Rainforest ecoregion. Our records reinforce the specific association between *Adoratopsylla* and *Polygenis* fleas and didelphid opossums, mentioned in the literature for the Atlantic Forest ecoregion in Brazil. Furthermore, we observed that flea communities in opossums change in a landscape gradient, with invasive cat fleas and euryxenous fleas common in anthropized areas, and endemic fleas common in natural areas. Our results underline the importance for public health and veterinary medicine the analysis of flea circulation between wild and urban environments due to the risk of pathogen transmission

Physiology

Blakely B.N., Agnew J., Gard C., Romero A.

Effects of blood meal source on blood consumption and reproductive success of cat fleas, *Ctenocephalides felis*.

(2023) PLoS Neglected Tropical Diseases, 17 (4), art. no. e0011233

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da Silva T.X.D.A., Almeida G.P.S., Correia T.R.

Monitoring the development of *Ctenocephalides felis felis* (Siphonaptera, Pulicidae) eggs from a laboratory colony in Brazil subjected to different adhesive substances.

(2023) Veterinary Parasitology: Regional Studies and Reports, 43, art. no. 100899

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Khokhlova I.S., Fielden L.J., Shenbrot G.I., Krasnov B.R.

Metabolic rate and ecological traits of ectoparasites: a case study with seven flea species from the Negev Desert.

(2023) Parasitology Research, 122 (10), pp. 2317 – 2324

We studied the relationship between fleas' metabolic rate and their ecological traits, using data on standard metabolic rate (SMR), mean abundance, host specificity, and geographic range size in males and females of seven desert flea species. SMR was measured via mass-specific CO₂ emission, whereas host specificity was measured as (a) the mean number of host species used by a flea per region

in regions where this flea was recorded; (b) the total number of host species a flea exploited across its geographic range; and (c) the phylogenetic diversity of the flea's hosts. To control for confounding effects of phylogeny when analysing data on multiple species, we applied the Phylogenetic Generalised Least Squares (PGLS) model. We found that the only ecological trait significantly correlating with flea SMR was the phylogenetic diversity of hosts utilized by a flea across its geographic range. The strength of the association between SMR and host phylogenetic diversity was higher in male than in female fleas. We explain the relationship between flea SMR and their host specificity by the necessity of host-opportunistic species to compensate for the high energetic cost of neutralizing multiple defences from multiple hosts by increased SMR.

[Variable metabolic energy costs of digesting blood may not only be due to defences of the hosts, but alternatively due to differences in the capabilities of the fleas. Particular digestive anatomies and enzymes (e.g. oxidative, proteolytic) may be needed to metabolize particular bloodmeals. Rothschild (1975) notes that digestion and the digestive systems of different flea species are themselves highly variable. Editor's note]

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Lu S., Danchenko M., Macaluso K.R., Ribeiro J.M.C.

Revisiting the sialome of the cat flea *Ctenocephalides felis*

(2023) PLoS ONE, 18 (1 January), art. no. e0279070

DOI: 10.1371/journal.pone.0279070

[The "sialome" refers in this paper to the molecules a salivary gland makes. Editor's note]

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Zduniak M., Serafini S., Wróbel A., Zwolak R.

Host body mass, not sex, affects ectoparasite loads in yellow-necked mouse *Apodemus flavicollis*.
(2023) Parasitology Research,

DOI: 10.1007/s00436-023-07958-5

Evolution

Liu Y., Chen B., Lu X., Liu S., Jiang D., Wang X., Yi L., Li R., Zhang Q., Wu L., Yang X.

Analysis of complete mitogenomes and phylogenetic relationships of *Frontopsylla spadix* and *Neopsylla specialis*.

(2023) Frontiers in Veterinary Science, 10, art. no. 1250381,

DOI: 10.3389/fvets.2023.1250381

Fleas represent a group of paramount medical significance, subsisting on blood and acting as vectors for an array of naturally occurring diseases. These pathogens constitute essential elements within the plague biome, exerting deleterious effects on both human and livestock health. In this study, we successfully assembled and sequenced the whole mitochondrial genome of *Frontopsylla spadix* and *Neopsylla specialis* using long-range PCR and next-generation sequencing technologies. The mitogenomes of *F. spadix* and *N. specialis* both have 37 genes with full lengths of 15,085 bp and 16,820 bp, respectively. The topology of the phylogenetic tree elucidates that species *F. spadix* is clustered in a branch alongside other members of the family Leptopsyllidae, whereas species *N. specialis* is a sister taxon to *Dorcadia ioffi* and *Hystrichopsylla weida qinlingensis*. It also suggests that Pulicidae form a monophyletic clade, Ctenophthalmidae, Hystrichopsyllidae, Vermipsyllidae form a sister group to Ceratophyllidae/Leptopsyllidae group. The mitochondrial genomes of *F. spadix* and *N. specialis* were sequenced for the first time, which will contribute to a more comprehensive phylogenetic analysis of the Siphonaptera order. The foundation for subsequent systematic studies, and molecular biology of fleas was established.

Ecology

Colombo V.C., Lareschi M., Monje L.D., Antoniazzi L.R., Morand S., Beldomenico P.M.

Ecological factors shaping the ectoparasite community assembly of the Azara's Grass Mouse, *Akodon azarae* (Rodentia: Cricetidae).

(2023) Parasitology Research, 122 (9), pp. 2011 - 2021,

DOI: 10.1007/s00436-023-07901-8

Parasites are integral members of the global biodiversity. They are useful indicators of environmental stress, food web structure and diversity. Ectoparasites have the potential to transmit vector-borne diseases of public health and veterinary importance and to play an important role in the regulation and evolution of host populations. The interlinkages between hosts, parasites and the environment are complex and challenging to study, leading to controversial results. Most previous studies have been focused on one or two parasite groups, while hosts are often co-infected by different taxa. The present study aims to assess the influence of environmental and host traits on the entire ectoparasite community composition of the rodent *Akodon azarae*. A total of 278 rodents were examined and mites (Mesostigmata), lice (Phthiraptera), ticks (Ixodida) and fleas (Siphonaptera) were determined. A multi-correspondence analysis was performed in order to analyze interactions within the ectoparasite community and the influence of environmental and host variables on this assembly. We found that environmental variables have a stronger influence on the composition of the ectoparasite community of *A. azarae* than the host variables analyzed. Minimum temperature was the most influential variable among the studied. In addition, we found evidence of agonistic and antagonistic interactions between ticks and mites, lice and fleas. The present study supports the hypothesis that minimum temperature plays a major role in the dynamics that shape the ectoparasite community of *A. azarae*, probably through both direct and indirect processes. This finding becomes particularly relevant in a climate change scenario.

**

Eads D.A., Huyvaert K.P., Biggins D.E.

Estimating parasite infrapopulation size given imperfect detection: Proof-of-concept with ectoparasitic fleas on prairie dogs.

(2023) International Journal for Parasitology: Parasites and Wildlife, 20, pp. 117 - 121,

DOI: 10.1016/j.ijppaw.2023.01.002

Parasite infrapopulation size - the population of parasites affecting a single host - is a central metric in parasitology. However, parasites are small and elusive such that imperfect detection is expected. Repeated sampling of parasites during primary sampling occasions (e.g., each host capture) informs the detection process. Here, we estimate flea (Siphonaptera) infrapopulation size on black-tailed prairie dogs (*Cynomys ludovicianus*, BTPDs) as a proof-of-concept for estimating parasite infrapopulations given imperfect detection. From Jun–Aug 2011, we live-trapped 299 BTPDs for a total of 573 captures on 20 plots distributed among 13 colonies at the Vermejo Park Ranch, New Mexico, USA. During each capture, an anesthetized BTPD was combed 3 times consecutively, 15 s each, to remove and count fleas. Each flea ($n = 4846$) was linked to the BTPD from which it was collected and assigned an encounter history ('100', '010', '001'). We analyzed the encounter histories using Huggins closed captures models, setting recapture probabilities to 0, thereby accounting for flea removal from hosts. The probability of detecting an individual flea (p) increased with Julian date; field personnel may have become more efficient at combing fleas as the field season progressed. Combined p across 3 combings equaled 0.99. Estimates of flea infrapopulation size were reasonable and followed the negative binomial distribution. Our general approach may be broadly applicable to estimating infrapopulation sizes for parasites. The utility of this approach increases as p declines but, if p is very low, inference is likely limited.

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Hanzelka J., Baroni D., Martikainen P., Eeva T., Laaksonen T.

Cavity-breeding birds create specific microhabitats for diverse arthropod communities in boreal forests.

(2023) Biodiversity and Conservation, 32 (12), pp. 3845 - 3874,

DOI: 10.1007/s10531-023-02653-4

The nests of secondary cavity-nesters located in tree cavities may form specific microhabitats of conservation importance due to their limited accessibility and availability. Species-specific nesting materials in nests of different secondary cavity-nesters may furthermore provide very different microhabitats for arthropods. The potential differences in arthropod communities inhabiting nests of different bird species in excavated cavities or nest boxes have, however, rarely been studied despite their relevance for conservation. Here we investigated the diversity and composition of arthropod communities in these different cavity types and bird species' nests in managed boreal forests. We identified morphologically and by DNA-metabarcoding arthropods in nest materials that were collected in and compared between (i) woodpecker-size cavities from seven different combinations of cavity type (nest box or excavated cavity), tree species (aspen or pine) and accumulation history of nest materials

(single-season cleaned or uncleaned nest boxes that accumulated nests of passerines or an owl species); and (ii) nests of two different passerine species in small nest boxes. We identified 64 arthropod taxa in ten orders, from which Diptera, Coleoptera, Siphonaptera, and Lepidoptera were the most abundant. Shannon diversity index was similar among the cavity-nest-type combinations, but taxa richness was the highest in the owl nests. The arthropod communities (especially Histeridae beetles) deviated most from the other types of nests in owl and aspen cavity nests with more advanced decomposition of nest material (soil or wet environment related taxa). The differences in arthropod communities between the different nest types point out the importance of the ecological chain “tree cavities—bird nests—arthropod communities”.

Pathology and Control

ACOSTA D.B, ZANOCCO F.A., RUIZ M., SANCHEZ J.P. 2023.

Domestic dogs as host of ectoparasites carrying *Rickettsia*, *Bartonella* and *Mycoplasma* in urban, peri-urban and rural areas from center Argentina.

Mastozoología Neotropical 31 (1) (in press).

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Boyde A., Mills D., Abba A.M., Ezquiaga M.C. (2023)

Fleas and lesions in armadillo osteoderms.

Journal of Anatomy, 242 (6), pp. 1029 – 1036. DOI: 10.1111/joa.13842

Microbial Symbiotes

Dewitte A., Werkmeister E., Pierre F., Sebbane F., Bontemps-Gallo S.

A Widefield Light Microscopy-Based Approach Provides Further Insights into the Colonization of the Flea Proventriculus by *Yersinia pestis*

(2023) Applied and Environmental Microbiology, 89 (4)

DOI: 10.1128/aem.02091-22

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Fukatsu T., Gottlieb Y., Duron O., Graf J.

Editorial: Microbial associates of blood-sucking arthropods and other animals: relevance to their physiology, ecology and evolution

(2023) Frontiers in Microbiology, 14, art. no. 1256275

DOI: 10.3389/fmicb.2023.1256275

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Kolodziejek A.M., Bearden S.W., Maes S., Montenieri J.M., Gage K.L., Hovde C.J., Minnich S.A.

Yersinia pestis Dail Mutants Are Not Susceptible to Human Complement Bactericidal Activity in the Flea.

(2023) Applied and Environmental Microbiology, 89 (2)

DOI: 10.1128/aem.01244-22

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Tan Y., Liu W., Chen Y., Zhou Y., Song K., Cao S., Zhang Y., Song Y., Deng H., Yang R., Du Z.

Comparative lysine acetylome analysis of *Y. pestis* YfiQ/CobB mutants reveals that acetylation of slyA lys73 significantly promotes biofilm formation of *Y. pestis*.

(2023) Microbiology Spectrum, 11 (4)

DOI: 10.1128/spectrum.00460-23

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Wu Y.-L., Hu S.-F., Zhang X.-L., Wang H.-M., Pan H.-Y., Liu G.-H., Deng Y.-P.

Complete bacterial profile and potential pathogens of cat fleas *Ctenocephalides felis*.

(2023) Acta Tropica, 243, art. no. 106923

DOI: 10.1016/j.actatropica.2023.106923

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Sensory perception of fleas, especially larvae.

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Flea taxonomy and host associations, especially in eastern North America and the Indo-Australian region.

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Host specificity of fleas.

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Taxonomy of fleas.

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 Taxonomy, host associations, and vectorial importance of fleas.

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 All groups of ectoparasites on vertebrates.

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 Flea systematics & taxonomy, host relationships, distribution & biogeography.

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 Taxonomy of fleas, host-parasite relationships, fleas as vectors of parasites.

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Ectoparasitic arthropods and their vectorial role.
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Ectoparasites of vertebrates, particularly mites.

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 Ectoparasite (especially flea) systematics, taxonomy, and ecology.

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Fleas in Art and History

The GOUT and the FLEA. A Fable.

By Anon. (William Starrat)
(born c. 1700, died c. 1768)

Anonymous poem in “Scotch Poems”, *The Ulster Miscellany*, 1753

When luxury and idleness
Did a’ the richer warld possess:
When careless nastiness, and dirt,
Was rife among the poorer sort:
Jove saw, and coost about betimes
For punishments for siccan crimes:
For he, foreby his thunderbouts,
Has rowth o’ plagues, as fleas — and gouts —
Thae same may do, quo’ Jove, faith I
Will send a sample out to try.

Forth came the flea, and eke the gout
While Jove his orders thus gave out:
Near hand a marble-biggings stands,
There lives a man wi’ flocks and lands;
And cross the lone a cotter dwells,
In thir twa houses six your sells;
Ilk ane his house — there ye maun bide,
But whilk in whilk, let fate decide.

They did their honours to the god,
And down Olympus took their road;
And trav’ling on, at length they saw
The houses that maun ha’d the twa.

The flea a greedy saul confess’d,
By hopping off to wale the best:
While the poor limping gout was forc’t,
To hobble hooly to the warst.

The flea hopp’d off, as we have said,
And wan into the rich man’s bed;
Himsel was there; when on his breast,
The weary trav’ller thought to feast:
It bit him sair, — the man haff mad,
Roar’d out wi’ a’ the strength he had,
Fy, bring a light — I’m out o’ breath,

There's something biting me to death.

They brought the light — the flea they saw,
 But nimble hoppy scap'd them a'.
 Again it ventures on his hips,
 Bites sair, and then the claret sips:
 The man, tho' dozing, felt the pain,
 Calls for the light — they hunt again,
 While wi' difficulty the flea,
 Darn'd in a lirk, in safety lay.

But the neist morn, the servants run,
 And hang the bedclaiiths in the sun;
 When hoppy like a man dismay'd,
 Gi'd twa 'r three lowps, and off he gaid.

There's many a man in hoppy's case,
 Who, struggling, gain some pompous place.

For naithing sorts wi' us, we see,
 But what the gods themselves decree.

The flea wi' danger thus turn'd out;
 Let's see what happen'd to the gout.

We left it in the little croove,
 Where it had hardly room to move;
 Sae thick the cogues and creepies lay,
 And in the nook a wad o' strae:
 There the poor man was streek'd at length,
 Wi' ease recruiting wasted strength.

In till the man, poor limpy crawls,
 And seiz'd upon his hinder spauls;
 The man unus'd to sic a guest,
 Could neither sleep, nor lie, nor rest.

Bless me, quo' he, what's this I feel?
 This maun be witchcraft, or the deel:
 That stounds me sae — then wi' a rair,
 (The gout afflicted him sae sair)
 He madly plung'd into a pool,
 The heat and tingling pain to cool:
 But what wi' water, rubs, and blows:
 The weary gout got sma' repose;
 And then neist morn, without remead,
 He maun gae dig for daily bread;
 Poor limpy cou'd na brook the spade,
 He dropp'd his leg, and off he gaid.

As ow'r the lone he bent his way,
He there foregather'd wi' the flea.

Then to ilk other they complain;
What does great Jove and mortals mean?
Why did great Jove our beings give,
When mortals winna let us live?
Here we maun bide — come let us try
A change of quarters e'er we die.

The gout gaid to the rich man's house,
Whare he liv'd happily and drowse:
There he gat ease, and flannels warm,
To keep him soft and free frae harm:
There he got morsels nice and fine,
And thrive fou fast, on drams, and wine.

Off to the cottage gaid the flea,
And nestled down in dirt and strae;
Attack'd the man, and bit him weel,
Sometimes he slept, and didna feel:
And when he did, he cou'dna grip it,
Sae nimbly throu' the strae it skipit.

Cosy it liv'd and e'er the week,
'Twas, like a coach horse, fat and sleek.
Ae sunday ev'ning, after mass,
The gout, and flea, like lad and lass,
Gaid to the place of rendezvous,
"How are you man? And how are you?
Trowth I'm as happy as a lord:
And I'm right happy tak my word."

Weel, quo' the gout, sin' baith are sae,
Let us e'en keep the beilds we hae;
Live happy in your humble state,
I'll bide amang the pamper'd Great.

La peste d'Elliant

By Louis Duveau

(b. 1818, St. Malo, France, d. 26 May 1867, Paris)

(1849, oil on canvas, 152x266 cm)



"Siphonaptera" Literature (2023)

[Editor's note: For this *Flea News*, I divided the literature for readability into 2 lists: "Siphonaptera" keyword and "Flea" keyword. The former list is focused on fleas, the latter list contains literature often only peripherally pertinent to fleas. There is overlap and repetition between the 2 lists.]

- Amarga A.K.S., Hastriter M.W. First record of bat flea *Thaumapsylla breviceps* Rothschild, 1907 (Siphonaptera: Ischnopsyllidae) on Bicol Region, Luzon Island, Philippines (2023) *Historia Naturalis Bulgarica*, 45 (5), pp. 103 - 106 DOI: 10.48027/hnb.45.051
- Amugune B.L., Matharu A.K., Ouma P., Mutebi F., Elson L., Fillinger U., Krücken J. Cost-Effective PCR-Based Identification of Tunga penetrans (Siphonaptera) Larvae Extracted from Soil Samples Containing PCR Inhibitor-Rich Material (2023) *Insects*, 14 (1), art. no. 5 DOI: 10.3390/insects14010005
- Backus L.H., López-Pérez A.M., Marcek J., Shultz L., Zazueta O.E., Shooter S., Foley J. Rickettsial antibodies and *Rickettsia bellii* detection in lagomorphs and their ectoparasites in Northern Baja California, Mexico (2023) *Journal of medical entomology*, 60 (5), pp. 1073 - 1080 DOI: 10.1093/jme/tjad085
- Baláž I., Bogdziewicz M., Dziemian-Zwolak S., Presti C.L.O., Wróbel A., Zduniak M., Zwolak R. From trees to fleas: masting indirectly affects flea abundance on a rodent host (2023) *Integrative Zoology*, 18 (3), pp. 440 - 452 DOI: 10.1111/1749-4877.12671
- Banda A., Gandiwa E., Muposhi V.K., Muboko N. Ecological interactions, local people awareness and practices on rodent-borne diseases in Africa: A review (2023) *Acta Tropica*, 238, art. no. 106743 DOI: 10.1016/j.actatropica.2022.106743
- Beliavskaia A., Tan K.-K., Sinha A., Husin N.A., Lim F.S., Loong S.K., Bell-Sakyi L., Carlow C.K.S., Abubakar S., Darby A.C., Makepeace B.L., Khoo J.J. Metagenomics of culture isolates and insect tissue illuminate the evolution of *Wolbachia*, *Rickettsia* and *Bartonella* symbionts in *Ctenocephalides* spp. fleas (2023) *Microbial Genomics*, 9 (7), art. no. 001045 DOI: 10.1099/mgen.0.001045
- Benkacimi L., Diarra A.Z., Bompar J.-M., Bérenger J.-M., Parola P. Microorganisms associated with hedgehog arthropods (2023) *Parasites and Vectors*, 16 (1), art. no. 211 DOI: 10.1186/s13071-023-05764-7
- Blakely B.N., Agnew J., Gard C., Romero A. Effects of blood meal source on blood consumption and reproductive success of cat fleas, *Ctenocephalides felis* (2023) *PLoS Neglected Tropical Diseases*, 17 (4), art. no. e0011233 DOI: 10.1371/journal.pntd.0011233
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